

5.2. Indication of at least 'one way'

An invention is in principle sufficiently disclosed if at least one way is clearly indicated enabling the person skilled in the art to carry out the invention. If this is the case, the non-availability of some particular variants of a functionally defined component feature of the invention is immaterial to sufficiency as long as there are suitable variants known to the skilled person through the disclosure or common general knowledge which provide the same effect for the invention (T 292/85, OJ 1989, 275). This has been confirmed by many decisions, for example: T 81/87 (OJ 1990, 250), T 301/87 (OJ 1990, 335), T 212/88 (OJ 1992, 28), T 238/88 (OJ 1992, 709), T 60/89 (OJ 1992, 268), T 182/89 (OJ 1991, 391), T 19/90 (OJ 1990, 476), T 740/90, T 456/91 and T 242/92.

If a claim comprises non-working embodiments, the consequences differ depending on the circumstances (G 1/03, OJ 2004, 413, point 2.5.2 of the Reasons, citing T 238/88, OJ 1992, 709; T 292/85, OJ 1989, 275, and T 301/87, OJ 1990, 335). This finding from G 1/03 is analysed in T 2210/16. See also T 875/16 (points 7 and 35 of the Reasons).

In T 941/16 (field of biochemistry) the board stated that the requirements of sufficiency of disclosure applied to all technical fields. Thus, depending on the individual case, inventions relating to a generic chemical formula comprising non-working embodiments with functional features contravened the requirements of Art. 83 EPC too.

In T 1809/17 the opponent argued that claim 1 was insufficiently disclosed in that it did not specify all essential features. For the board, however, the crucial point was not whether the independent claim defined all essential features but rather whether the patent specification as a whole provided the skilled person with at least one workable example of how the claimed invention could be implemented.

5.3. Examples

Whether or not the disclosure of the patent in suit is sufficiently clear and complete within the meaning of Art. 100(b) and 83 EPC must be decided by appraising the information contained in the examples as well as other parts of the description in the light of the common general knowledge of the skilled person at the priority date (T 322/93 and T 524/01).

However, where the application disclosed the claimed invention in a manner sufficiently clear and complete for it to be carried out by a person skilled in the art, it then necessarily disclosed at least one way of carrying out the invention disclosed in accordance with R. 42(1)(e) EPC, with the necessary details being derivable from the description including the prior art referred to therein (see e.g. T 389/87, T 561/96 and T 990/07). The board in T 990/07 pointed out that, although the case underlying decision T 561/96 differed from that in T 990/07 in that in T 561/96 the description and the drawings were not erroneous, the board in T 561/96 had also held that in cases where examples were not indispensable, their omission did not contravene R. 27(1)(e) EPC 1973 (R. 42(1)(e) EPC). This Rule only required the insertion of such examples "where appropriate". The jurisprudence of the boards of appeal thus drew a clear distinction between the concepts of "way of carrying

out the invention claimed" and "examples" referred to in R. 27(1)(e) EPC 1973. According to this jurisprudence, the detailed description of one way of carrying out the invention claimed had to be interpreted in the light of Art. 83 EPC. It constituted a condition to be met by the description as a whole and was clearly mandatory. In contrast, the presence of examples would only be indispensable if the description would otherwise not be sufficient to meet this requirement. Hence, the purpose of the "examples" evoked in R. 27(1)(e) EPC 1973 appeared primarily to be to complete an otherwise incomplete teaching. See also T. 1918/07 and T. 1169/08.

In T. 226/85 (OJ 1988, 336), T. 409/91 (OJ 1994, 653) and T. 694/92 (OJ 1997, 408) the patent or the patent application disclosed only one or very few ways of carrying out the invention. In each of these decisions the boards had held that the disclosure of the specific examples was not sufficient to enable the invention to be carried out as claimed. However, according to T. 617/07, no principle could be deduced from these decisions that sufficiency of disclosure is always to be denied if there is only example of carrying out an invention. Rather all three decisions emphasised that an objection of lack of sufficiency of disclosure (i) presupposes that there are serious doubts, substantiated by verifiable facts, and (ii) depends on the evidence available in each case whether or not a claimed invention can be considered as enabled on the basis of the disclosure of one worked example.

In T. 721/16 the patent was revoked by the opposition division. The objection of lack of sufficiency of disclosure related to the measurement of parameters defining the polyvinylpyrrolidone powder compositions. While it was not disputed that the parameters defined in claims 1, 2 and 3 were generally known to the skilled person and that they were also defined in the patent in suit, the respondent (opponent) argued that the patent in suit did not disclose all the necessary conditions to perform reproducible measurements of these parameters. The arguments of the respondent in T. 721/16 concerned in fact the repetition of the examples of the patent in suit, i.e. the preparation of polyvinylpyrrolidone powder compositions having specific K values, K value lowering ratios and content of insoluble substances. However, the board stated that the difficulty in providing an exact repetition of the examples of the patent in suit was in the case in hand not decisive for concluding a lack of sufficiency of disclosure of the claimed invention and could be left unanswered. Firstly, it is the sufficiency of disclosure of the combination of technical features of the invention, i.e. as defined by the terms of the claims (see R. 43(1) EPC), which has to be assessed and not that of the specific exemplified embodiments, which were not in the case in hand the subject-matter of a claim. Secondly, according to R. 42(1)(e) EPC the description shall describe in detail at least one way of carrying out the invention claimed, using examples where appropriate, meaning that the presence of examples is not a mandatory requirement for meeting the requirement of sufficiency of disclosure. (On this last point, see also T. 1437/07, point 38 ff of the Reasons.)

In T. 2242/16 the appellant (patent proprietor) took the view, based on T. 665/90, that proving the alleged lack of enablement would have required reproducing an example of the patent in exactly the same way. The board decided that there could be no such requirement in the case in hand, finding that, at least according to the patent's teaching, the precise approach taken in the example was not crucial. The board referred to a number

of other decisions (T.740/90, T.406/91, T.1712/09) and to the established case law reported in chapter II.C.5.4.

5.4. Invention to be performed over whole range claimed

Under the established case law of the boards of appeal, the requirements of sufficiency of disclosure are met if a person skilled in the art can carry out the invention as defined in the independent claims over the whole scope of the claims without undue burden using their common general knowledge (e.g. T.409/91, OJ 1994, 653; T.435/91, OJ 1995, 188).

The disclosure of one way of performing an invention is only sufficient if it allows the invention to be performed **in the whole range claimed** rather than **only in some** members of the claimed class to be obtained (T.409/91, OJ 1994, 653; T.435/91, OJ 1995, 188; and T.172/99). This is considered a question of fact. Sufficiency of disclosure thus presupposes that the skilled person is able to obtain substantially **all** embodiments falling within the ambit of the claims. This view has been taken by the board in numerous decisions, for example T.19/90 (OJ 1990, 476), T.418/91, T.548/91, T.659/93, T.435/91 (OJ 1995, 188) and T.923/92 (OJ 1996, 564; even more recently, issue discussed in detail in T.1727/12 ("Biogen Sufficiency")). This principle applies to any invention irrespective of the way in which it is defined, be it by way of a functional feature or not. The peculiarity of the functional definition of a technical feature resides in the fact that it is defined by means of its effect. That mode of definition comprises an indefinite and abstract host of possible alternatives, which is acceptable as long as all alternatives are available and achieve the desired result; it therefore has to be established whether or not the patent discloses a technical **concept fit for generalisation** which makes the host of variants encompassed by the functional definition available to the skilled person (T.1121/03 (no technical concept fit for generalisation – undue burden in carrying out the invention over the whole scope claimed – research programme) and T.369/05; see also T.2128/13). With respect to a claimed process defined in a functional manner, i.e. by its outcome, the board in T.1051/09 concluded that what was lacking was a generalisable teaching applicable within the scope of the claims, i.e. beyond the specific examples.

More technical details and more than one example may be necessary in order to support claims of a broad scope (T.612/92, T.694/92, OJ 1997, 408; T.187/93). This must be decided on a case-by-case basis. The board must also be satisfied firstly that the patent specification put the skilled person in possession of at least one way of putting the claimed invention into practice, and secondly that the skilled person could put the invention into practice over the whole scope of the claim. If the board was not satisfied on the first point that one way existed, the second point did not need to be considered (T.792/00).

According to the decision in T.517/98 (grains of superconductor material – density) the patent in suit had only shown the way of achieving the goal to some extent and within set limits, but certainly not over the whole ambit of the claim. In the board's judgement, broad protection such as that claimed was therefore unjustified since such a claim encompassed **speculative subject-matter** which could not be produced by the skilled person without inventive ingenuity (see also T.409/91). The main request failed (Art. 83 EPC).

In T.1064/15, the invention related to barbed suture-needle combinations useful for connecting body tissue in various surgical contexts. Claim 1's requirements could be implemented for circular cross-sections, but the question of sufficiency of disclosure arose for non-circular cross-sections. The respondent (patent proprietor) himself intended the teaching of the patent to be applicable to both circular and non-circular cross-sections and specifically sought protection for both types of embodiment. It would be insufficient and disproportionate if the sole disclosed possibility of carrying out the invention with circular cross-section elongated bodies were enough to satisfy the requirements of sufficiency of disclosure. Such an approach could not have been intended by the legislator, because in the board's view this would go against the general principle that the protection obtained with the patent had to be commensurate with the disclosed teaching. When it came to non-circular cross-sections, this was not the case for the patent in suit. On the basis of the patent disclosure as a whole, taking common general knowledge into account, the person skilled in the art was not able to determine which dimension was meant by the diameter (SD) for an essential part of the claim, or in other words, with a needle having a given diameter, he did not know how to select the cross-section dimension of a non-circular suture in order to improve the closure strength, which was supposed to be an essential part of the teaching of the patent in suit (T.1064/15, which is cited and summarised in T.1756/16).

In T.713/15 the patent related to a novel preventive and/or therapeutic agent (anti IL-6 receptor antibody) for vasculitis in general and did not distinguish between various types of vasculitis. There was no teaching in the application as filed on how to treat vasculitis in all its forms (like Behçet's disease). The subject-matter of claim 1 of the main request was therefore not sufficiently disclosed over the whole scope of the claim. The patent proprietor had filed an expert declaration based on clinical data, but it did not address D(49) and D(50) submitted by the opponent and so could not provide crucial input.

In T.553/10, the board stated that the passages cited by the appellant disclosed a method for producing lithium nickel manganese cobalt oxides which fell either within or outside the ambit of claim 1. An additional process step required when seeking to prepare oxides falling within the ambit of claim 1 was missing. The application lacked guidance, and this could not be overcome by drawing on common general knowledge. A declaration written by **an employee** of the appellant was therefore of little probative value for establishing what was common general knowledge in the art.

In T.239/13 of 5 July 2017, claim 1 as granted did not require the "granules" to be acidic. The board considered that a solution of the claimed granules per se might have an alkaline pH despite the presence of some acidic component. As regards acidic granules, the description – which dealt exclusively with acidic granules – provided the skilled person with technical information and guidance sufficient to enable him to prepare, without undue burden, acidic granules having "improved storage properties" across the whole ambit of claim 1. As regards alkaline granules, in the absence of a concrete teaching the skilled person would have to start a research programme. The board concluded that the skilled person, following the teaching of the description, was not provided with technical information and guidance sufficient to enable him to prepare granules as claimed without undue burden and across the whole ambit of claim 1.

In T 1994/12 (rubber composition), as to the argument that the requirement for sufficiency of disclosure was not met because the skilled person in view of the ambiguity in respect of the nature of the asphalt would not be able to reproduce the examples of the patent, the board stated that sufficiency of disclosure was not concerned with the invention **the applicant might have had in mind** when drafting the application, but rather with the invention defined by the claims in terms of the technical features of the invention (see R. 43(1) EPC), as is done for assessing other criteria for patentability such as novelty and inventive step.

In T 116/18 (OJ 2022, A76) synergism was not a feature of the claims of the patent in suit. Consequently, whether this effect was achieved over the entire breadth of the claims was not to be assessed under Art. 100(b) EPC but Art. 56 EPC (G 1/03 (OJ 2004, 413), point 2.5.2 of the Reasons). Case T 116/18 is a referral pending under G 2/21.

In T 623/16 (OLEDs) the description proposed varying a **plurality of parameters** to achieve a white emitting device. There was no indication in the claims that several parameters, such as concentration and thickness on the one hand and the addition of a blocking layer (or other parameters) on the other, should be varied at the same time. If the skilled person were to make a first experiment and obtain a device which did not emit white, they would find no explicit guidance in the patent for deciding on what to start with to yield white emission. Thus, guidance had to be given in the claims as to which features were essential for carrying out the invention over the whole range in order to achieve a device emitting white light. Once the skilled person was aware that these features were crucial for carrying out the invention, changing only concentration and thicknesses was no longer an undue burden. The description provided enough details on carrying out the invention. Claim 1 of the main request did not comprise features that were essential to the invention and crucial to carry it out over the whole range claimed. Omitting these features led to a device that did not emit white light (Art. 83 EPC not complied with). With amendments introducing the essential features, it was no longer an undue burden for the skilled person (new first auxiliary request satisfied Art. 83 EPC).

In T 2773/18 (mechanics) the patent concerned a wind turbine with a cooling device using outside air, in particular for marine environments. In support of an alleged lack of sufficient disclosure concerning the expression "upper part of the tower", the appellant (opponent) argued that not only was the claim was not limited to an offshore wind turbine but the dimension of the lower and upper parts was also not limited to a minimum size or height. The scope of claim 1 thus covered embodiments where the inlet was located quite low above sea level and therefore unable to achieve the technical effect of drawing outside air with a low water and salt content. This argument failed to convince the board, not least because it misapplied **case law developed in the field of chemistry** – for where a claimed invention was in a compositional range or other range of values but the associated effect was perhaps not proven or plausible for large parts of that range – to an invention claimed in the field of mechanics, even though it claimed no ranges. By its very nature, a claim in the field of mechanics, which – often in functional or other generic terms – attempted to capture the essence of some specific machine or mechanical structure (or its operation), was schematic and allowed for some breadth of interpretation. It might be that, on clever construction, subject-matter could be found to be covered by that breadth that